AI-Based Autonomous Multi-Sensor Search and Rescue Drone for Survivor Localization in Landslide Areas

July 11, 2026

ABSTRACT

Natural disasters such as landslides, floods, and earthquakes often delay rescue operations because victims are difficult to locate in dangerous environments. Traditional search methods require significant manpower and expose rescue personnel to additional risks. This project proposes a low-cost AI-Based Autonomous Multi-Sensor Search and Rescue Drone capable of assisting rescue teams during disaster response.

The proposed prototype integrates a Pixhawk flight controller, ESP32 microcontroller, ESP32-CAM, 24 GHz mmWave radar, GPS module, microphone, and BME280 environmental sensor. Instead of performing Artificial Intelligence processing onboard the drone, all collected sensor data is transmitted wirelessly to a laptop where AI-based computer vision and sensor fusion algorithms estimate the probability of survivor presence.

The proposed system provides GPS coordinates of suspected survivor locations and displays them on a rescue dashboard. The modular architecture allows future upgrades using UWB Life Detection Radar, LiDAR, and onboard AI computers without redesigning the complete system.

1 Introduction

Landslides are among the most destructive natural disasters, causing significant loss of life and property every year. Rescue teams often encounter unstable terrain, collapsed structures, and poor visibility, making victim localization extremely difficult.

Recent developments in Unmanned Aerial Vehicles (UAVs), Artificial Intelligence (AI), and embedded systems have created new opportunities for improving disaster response. Autonomous drones can rapidly survey hazardous areas while minimizing risks to rescue personnel.

This project proposes a low-cost prototype capable of collecting visual and environmental information using multiple sensors. The collected data is transmitted to a laptop where AI algorithms analyze the information to assist rescue teams in locating possible survivors.

2 Problem Statement

Conventional rescue operations rely mainly on manual searching, rescue dogs, and helicopters. These approaches require considerable time and may expose rescue personnel to dangerous environments.

Most affordable drones only provide aerial images and lack intelligent analysis capabilities. Furthermore, many low-cost sensors cannot independently confirm the presence of survivors, resulting in false alarms.

Therefore, there is a need for an affordable autonomous drone capable of collecting information from multiple sensors and combining the data using Artificial Intelligence to improve survivor localization.

3 Objectives

The objectives of the proposed system are:

- Design a low-cost autonomous search and rescue drone.
- Capture real-time images using ESP32-CAM.
- Detect human presence using a 24 GHz mmWave radar.
- Record GPS coordinates of suspected survivor locations.
- Detect cries or tapping sounds using a microphone.
- Monitor environmental conditions using the BME280 sensor.
- Transmit all sensor data to a laptop.
- Perform AI-based human detection using computer vision.
- Implement sensor fusion to improve detection reliability.
- Display survivor probability and GPS location on a rescue dashboard.

4 Literature Survey

Several research studies have proposed sensor-based systems for locating disaster victims. Most systems use ultrasonic sensors, vibration sensors, CO_2 sensors, microphones, or environmental sensors. Although these approaches reduce equipment cost, they cannot reliably detect survivors buried beneath dense landslide debris.

Other studies combine Artificial Intelligence with multiple sensors to reduce false alarms. However, many existing systems do not integrate autonomous drones, GPS mapping, or real-time decision support.

Based on the literature survey, the following research gaps were identified:

- Lack of autonomous aerial surveying.
- Limited use of Artificial Intelligence.
- Absence of sensor fusion algorithms.
- Limited GPS-based victim localization.
- Poor scalability for future upgrades.

The proposed project addresses these limitations by integrating autonomous flight, AI-based image processing, and multi-sensor data fusion within a low-cost prototype.

5 Proposed System

The proposed system is a low-cost autonomous search and rescue drone that combines multiple sensors with Artificial Intelligence to assist rescue teams during landslides and other natural disasters. Instead of processing Artificial Intelligence algorithms onboard the drone, the system transmits all collected sensor data to a laptop where computer vision and sensor fusion algorithms estimate the probability of survivor presence.

The drone performs autonomous flight using the Pixhawk flight controller while the ESP32 microcontroller collects data from all sensors. The ESP32-CAM captures real-time images and transmits them together with sensor information to the ground station. The laptop processes the incoming information using Python, OpenCV and YOLO for human detection.

The modular architecture allows future upgrades such as UWB Life Detection Radar, LiDAR, onboard AI processors, and long-range communication without changing the overall system design.

6 System Architecture

The proposed architecture consists of two major subsystems.

6.1 Drone Subsystem

The drone subsystem is responsible for collecting data.

It consists of:

- Pixhawk Flight Controller
- ESP32 Microcontroller
- ESP32-CAM
- 24 GHz mmWave Radar
- GPS Module
- Microphone Module
- BME280 Environmental Sensor
- Telemetry Module
- LiPo Battery

6.2 Ground Station

The ground station consists of a laptop running Python software.

The software performs

- Image Processing
- Human Detection
- Sensor Fusion
- GPS Mapping
- Rescue Dashboard

7 Block Diagram

Sensors (Radar, GPS, Microphone, BME280)

↓

ESP32 Controller

↓

ESP32-CAM

↓

Wi-Fi / Telemetry

↓

Laptop AI Software

↓

Sensor Fusion + Dashboard

8 Hardware Components

Component	**Function**
Pixhawk	Controls autonomous flight and stabilization.
ESP32	Reads sensor data and transmits it to the laptop.
ESP32-CAM	Captures live images for AI-based human detection.
24 GHz Radar	Detects human presence and movement.
GPS Module	Provides real-time location coordinates.
Microphone	Detects cries or tapping sounds.
BME280	Measures temperature, humidity and air pressure.
Telemetry Module	Enables wireless communication.
LiPo Battery	Supplies power to the drone.

Table 1: Hardware Components

9 Working Principle

The drone follows predefined GPS waypoints using the Pixhawk flight controller. During flight, the ESP32 continuously reads information from the GPS module, 24 GHz mmWave radar, microphone and BME280 sensor while the ESP32-CAM captures live images.

All sensor data and camera frames are transmitted to the ground station through Wi-Fi or telemetry. The laptop software performs AI-based human detection using YOLO and combines all sensor information using a sensor fusion algorithm.

If the calculated survivor probability exceeds a predefined threshold, the software records the GPS location and displays an alert on the rescue dashboard.

10 Sensor Fusion

Instead of relying on a single sensor, the proposed system combines information from multiple sources.

The AI software analyses

- Camera Images
- Human Presence Radar
- Microphone Signals
- Environmental Conditions
- GPS Coordinates

This multi-sensor approach reduces false alarms and improves decision-making by estimating a confidence score for each detected location.

11 Estimated Budget

Component	**Cost (INR)**
F450 Frame	2000
Brushless Motors	3200
ESC	2000
Pixhawk	9000
GPS + Telemetry	4000
ESP32	600
ESP32-CAM	900
24 GHz Radar	2000
Microphone	250
BME280	400
Battery + Charger	6000
Miscellaneous	1500
Estimated Total	**31,850**

12 Future Scope

The proposed prototype is designed with a modular architecture that supports future upgrades. Advanced technologies such as UWB Life Detection Radar, LiDAR, onboard AI processors, satellite communication and autonomous path planning can be integrated without redesigning the complete system.

13 Conclusion

This project presents a low-cost AI-based autonomous search and rescue drone capable of assisting rescue teams during disaster response. The system combines autonomous flight, computer vision and multi-sensor data collection to identify possible survivor locations and provide GPS coordinates for rescue operations. The modular design makes the prototype suitable for future enhancements and real-world deployment after further research and funding.